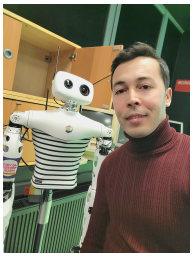
Yassine OUZAR

Assistant Professor
Université Paris 8

2 rue de la Liberté, 93526 Saint-Denis
Contact: +33 7 51 10 36 00
yassine.ouzar@univ-paris8.fr
My Webpage
Linkedin *Google Scholar*



Experience

2025-ongoing **Assistant Professor**, Université Paris 8, Saint-Denis, France

Research : LIASD

Teaching : UFR MITSIC

Teaching and research interests: artificial intelligence, computer vision, image/signal processing, affective computing, biometry, biomedical engineering

2024 - 2025 **Postdoctoral Researcher**, Lille Neurosciences & Cognition (LilNCog), INSERM, CRIStAL Lab, Université de Lille, Lille, France

Research topic: Identification of objective biomarkers of psychiatric disorders through multimodal analysis of physiological and behavioral data

2023 - 2024 **Postdoctoral Researcher**, CRIStAL Lab, CHU de Lille, Université de Lille, Lille, France

Research topic: Interpretable deep anomaly detection for the detection of medication prescribing errors

2022 - 2023 **Research and Teaching Assistant,**, Université de Lorraine, Metz, France.

Teaching (UFR SciFA) : Algorithms and Programming, Digital Electronics, GUI, OOP, IoT, Web Dev, UNIX, Computer Network, ML

Research (LCOMS) : remote vital signs measurement, physio-visual fusion, affective computing

Education

2023 **PhD**, *Université de Lorraine*, Metz, France, Contactless affective state recognition via physio-visual fusion from facial video.
See **PhD Thesis section**.

2019 **Master**, *Université Paris Saclay*, Orsay, France, Master of Embedding Systems and Information Processing.
Internship: ETIS Lab. Research : Imitation learning and computer vision applied to secondary facial expression recognition.

2018 **Master**, *Université Abdelhamid Ibn Badis*, Mostaganem, Algérie, Master in Embedded Systems Electronics..
Internship: LEOG Lab. Research : Design and development of a hardware and software platform for a remote laboratory dedicated to electronics practical work.

2016 **Bachelor**, *Université Abdelhamid Ibn Badis*, Mostaganem, Algérie.
Bachelor of Electronic Engineering

PhD Thesis

title Video-based affective state recognition using physio-visual fusion

supervisors Choubeila Maaoui and Frederic Bousefsaf

examiners Yannick Benezeth, Frédéric Vanderhaegen, Samia Bouchafa and Bruno Emile

Research Activities

Research interests

computer vision, signal processing, multimodal fusion, biomedical engineering, affective computing, biometry.

Mentorship

PhD students — 3 ongoing theses (co-supervision)

Houari Fellah, Heart attack prediction using deep learning based on acoustic analysis of heart rhythm.

Yazid Moussaoui, Design and deployment of artificial intelligence agents and large language models for automation and remediation in enterprise cybersecurity.

Said Si kaddour, Multimodal and Soft Biometrics for Smart Digital City Applications.

Master's internships — 5 students : Lynda Lagha, Lydia Ouhib, David Lin, Chenxi Yang, Oussama Berhili

Projects

Research AAP — Bio-inspired vision: From cognitive multimodality to multitask learning. Funding: 5600 euros

Challenges

V4V — Participation to the 1st edition (2021). Rank (observed metric: RMSE) : 3 over 6.

Publications

International Journals (Indexed with Impact Factor)

CBM 2025 — **Y. Ouzar**, F. Ajmi, S. Ben Othman, C. Rousseliere, B. Decaudin, P. Odou, and S. Hammadi. Interpretable One-Class Classification Framework for Prescription Error Detection Using BERT Embeddings and Dimensionality Reduction. Computers in Biology and Medicine.

JAD 2025 — A. Amad, **Y. Ouzar**, C. Nineuil, F. Boutaleb, E. Pierson, F. D'Hondt, and M. Daoudi. Peripheral skin temperature differentiates unipolar and bipolar depression. Journal of affective disorders.

CBM 2023 — **Y. Ouzar**, D. Djeldjli, F. Bousefsaf, and C. Maaoui. X-iPPGNet: A novel one stage deep learning architecture based on depthwise separable convolutions for video-based pulse rate estimation. Computers in Biology and Medicine.

BSPC 2022 — F. Bousefsaf, T. Desquins, D. Djeldjli, **Y. Ouzar**, C. Maaoui, and A. Pruski. Estimation of blood pressure waveform from facial video using a deep U-shaped network and the wavelet representation of imaging photoplethysmographic signals. Biomedical Signal Processing and Control.

CBM 2021 — F. Bousefsaf, D. Djeldjli, **Y. Ouzar**, C. Maaoui, and A. Pruski. iPPG 2 cPPG: reconstructing contact from imaging photoplethysmographic signals using U-Net architectures. Computers in Biology and Medicine.

Peer-Reviewed International Conferences and workshops

BioSMART 2026 — L. Ouhib, **Y. Ouzar**, Z. Pinseel, S. Bouilland, and M. Ammi. Towards Objective Dysgraphia Detection: A Multi-Branch Deep Learning Approach for Online Handwriting Analysis. **(Accepted)**

FG 2025 — **Y. Ouzar**, C. Nineuil, F. Boutaleb, E. Pierson, A. Amad, and M. Daoudi. Wearable-Derived Behavioral and Physiological Biomarkers for Classifying Unipolar and Bipolar Depression Severity. IEEE International Conference on Automatic Face and Gesture Recognition.

ICPRW 2022 — **Y. Ouzar**, L. Lagha, F. Bousefsaf, and C. Maaoui. Multimodal stress state detection from facial videos using physiological signals and facial features. International Conference on Pattern Recognition Workshops.

CVPRW 2022 **Y. Ouzar**, D. Djeldjli, F. Bousefsaf, and C. Maaoui. Video-based multimodal spontaneous emotion recognition using facial expressions and physiological signals. IEEE/CVF International Conference on Computer Vision and Pattern Recognition Workshops.

ICCVW 2021 **Y. Ouzar**, D. Djeldjli, F. Bousefsaf, and C. Maaoui. LCOMS Lab's approach to the Vision for Vitals (V4V) Challenge. IEEE/CVF International Conference on Computer Vision Workshops.

Peer-Reviewed French Conferences

IC 2026 L. Ouhib, **Y. Ouzar**, Z. Pinseel, S. Bouilland, and M. Ammi. Vers une détection objective de la dysgraphie : une approche d'apprentissage profond multi-branches pour l'analyse en ligne de l'écriture manuscrite. **(Submitted)**

GRETSI 2022 F. Bousefsaf, D. Djeldjli, **Y. Ouzar**, C. Maaoui, and A. Pruski. Transformée en ondelettes et IA pour la reconstruction d'un signal PPG en contact à partir de sa version sans contact. XXVIIIème Colloque Francophone de Traitement du Signal et des Images, Nancy, Nov. 2022.

HANDICAP 2022 F. Bousefsaf, T. Desquins, D. Djeldjli, **Y. Ouzar**, C. Maaoui, and A. Pruski. Estimation sans contact de la tension artérielle par intelligence artificielle. Conférence Handicap, Paris, June 2022.

IFRATH 2021 **Y. Ouzar**, F. Bousefsaf, and C. Maaoui. Mesure sans contact de la fréquence cardiaque par caméra basée sur l'apprentissage profond. Colloque Jeunes Chercheurs IFRATH, Paris, Oct. 2021.

Conferences without Peer-Review

SAGIP 2022 **Y. Ouzar**, F. Bousefsaf, D. Djeldjli, and C. Maaoui. Reconnaissance multimodale des émotions spontanées par caméra basée sur les expressions faciales et les signaux physiologiques. Journées de printemps de la SAGIP 2022, Bidart, May 2022.

STP 2020 **Y. Ouzar**, F. Bousefsaf, C. Maaoui, and K. Chelghoum. Système bimodal pour la reconnaissance des émotions basé sur l'apprentissage profond. 27e Journées STP du GdR MACS, Nantes, Feb. 2020.

Teaching Activities

Area	Computer and Engineering Science, Artificial Intelligence
sectors	Licence and Master
Licence	Algorithms and Programming, Digital Electronics, GUI, Python programming
Master	UNIX, Computer Network, Web Developement, OOP, IoT, Open-source Software, Machine Learning, Big Data Indexing, Uncertainty Management, Explainable AI, Biometry

Services and Reviewing

Program Committee	2026: IEEE FG
	2025: DL4BA, DL4BBDA and MLESP Workshops
Reviewer	2026: ACM TOMM, Empathic AI Workshop
	2025: IEEE FG, IEEE JBHI (x2)
	2024: IEEE TMC, Sensors (x2)
	2023: ICCSEA